Qn. Set Code-7

Semester: 5th
Programme: B.Tech
Branch: CSE, E&CSE

AUTUMN END SEMESTER EXAMINATION-2024

5th Semester B.Tech

DISTRIBUTED OPERATING SYSTEMS

CS30009

(For 2023 (L.E), 2022 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) Write two advantages of distributed systems over centralized ones.

(b) Concurrency transparency is a desirable goal for distributed systems. Do centralized systems have this property automatically? Justify your answer.

(c) Why the architecture of Asynchronous Transfer Mode (ATM) Networks are called as cell-based architecture?

(d) Explain the concept of "mailbox" in buffered primitives.

(e) Explain the terms "clock tick" and "clock skew" used in logical clocks.

(f) Why deadlock avoidance is never used in distributed system?

(g) Explain intermittent fault and permanent fault in distributed system.

(h) What is active replication for fault tolerance in distributed systems?

(i) Explain the concept of "Memnet" in ring-based multiprocessor system.

(j) In worst case how many messages are required to select a co-ordinator in Bully algorithm.

SECTION-B

2. (a) Construct 8 × 8 omega network for communication between 8 computers and 8 memory locations and calculate the following: [5]

i. Total number of stages used in the network.

ii. Total number of switches required in the network.

iii. Total number of switches required in every stage of the network.

(b) Explain all the 14 steps used in the Remote Procedure Call from Client-to-Server. [5]

3. (a) What is transparency design issue in distributed system? Explain location transparency, and migration transparency. [5]

(b) Suppose that the time to do a null RPC (i.e., 0 data bytes) is 1.0 msec, with an additional 1.5 msec for every 1K of data. How long does it take to read 32K from the file server in a single 32K RPC? How about as 32 1K RPCs? [5]

4. (a) Consider the behaviour of two machines in a distributed system. Both have clocks that are supposed to tick 1000 times per millisecond. One of them actually does, but [5]

the other ticks only 990 times per millisecond. If UTC updates come in once a minute, what is the maximum clock skew that will occur?

(b) A distributed system may have multiple, independent critical regions. Imagine that process 0 wants to enter critical region A and process 1 wants to enter critical region B. Can Ricart and Agrawal's algorithm (distributed algorithm for mutual exclusion) lead to deadlocks? Justify your answer. [5]

5. (a) Explain the situation of "false deadlock" that occurs in the centralized deadlock detection algorithm. Explain how you can overcome this situation using "Chandy-Misra-Haas algorithm" for distributed deadlock detection. [5]

(b) Differentiate between distributed deadlock detection and distributed deadlock prevention. A process with transaction timestamp 50 needs a resource held by a process with transaction timestamp 100. Compare the results when: [5]

i. Wait-die deadlock prevention algorithm is used.

ii. Wound-wait deadlock prevention algorithm is used.

6. (a) In a certain workstation based system, the workstations have local disks that hold the system binaries. When a new binary is released it is sent to each workstation. Some workstations may be down when this happens. Design an algorithm that allows the updating to be done automatically, even though workstations are occasionally down. [5]

(b) Describe "fault tolerance system" in distributed environment. Explain the concept of fault tolerance using: [5]

i. Active replication.

ii. Primary backup.

7. (a) Explain Bully algorithm and Ring algorithm with diagrams. [5]

(b) Explain any two strong and weak Data-Centric Consistency models with examples for each of them. [5]

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